Seminar: Enterprise AI and Urban Analytics

Seminar Paper

Classical NLP for Sentiment and Aspect-Based Analysis of Amazon Reviews

A Study of the Trade-off Between Predictive Performance and Interpretability

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Abstract

Online reviews pair a star rating with free text that explains it, offering a continuous, survey-free feedback signal — if the text can be read at scale. This paper builds an end-to-end classical NLP pipeline (no large language models) that turns 87,713 Amazon reviews from two subscription categories into structured opinion data, and studies the central trade-off between predictive performance and interpretability. For document-level polarity (RQ1) we compare a majority-class baseline against Naive Bayes, Logistic Regression and gradient boosting on TF-IDF features, under binary and ternary framings and with explicit class-imbalance handling. Feature representation, imbalance handling and hyperparameters are selected on a held-out validation slice; final-test *labels, predictions and performance metrics* are never used for model or hyperparameter selection (the descriptive exploratory analysis in Section 3 runs on the full dataset, including the test period, as is standard practice for EDA). Bootstrap confidence intervals and a matched-sample comparison give every model family identical training data. Logistic Regression with balanced class weights is the **strongest observed model under this evaluated setup** (binary macro-F1 0.888) without sacrificing feature-level interpretability, including in the matched-sample comparison; the headline accuracy of a trivial baseline (69% on the final test slice) is shown to be misleading because it never identifies a single negative review. A ternary framing exposes **persistent difficulty under the evaluated setup**: the minority 3-star (neutral) class collapses to below 1% recall for Naive Bayes without rebalancing, and a dedicated automatic analysis of true 3-star test reviews shows the model cannot reliably separate mildly-positive from mildly-negative phrasing. For aspect-based sentiment (RQ2) we segment reviews into sentences, detect eight aspects with a cleaned keyword lexicon, use noun phrases for exploratory candidate discovery, and score each aspect-mentioning sentence with VADER — a lexicon scorer chosen because the document classifier is out-of-distribution on single sentences. The lexicon deliberately excludes broad product identifiers such as *box*, *magazine* and *subscription*, and removes sentiment words from aspect detection. Across both categories, *billing* (cancel/refund/charged) is the **weakest high-coverage aspect (mention rate $\geq 5\%$) in both categories**, while *content*, *quality* and *price* are the most positive high-coverage aspects. Billing-related complaint language is frequent and scores negative under the lexicon, especially for Subscription Boxes, and a clause-level sensitivity analysis reinforces the direction of this signal; however, a billing-lexicon masking check shows the polarity is **not robust for Subscription Boxes**: masking the triggering terms themselves shifts the mean sentiment from -0.080 to $+0.044$, reversing its sign, so part of the unmasked negativity reflects the lexicon valence of words like *cancel* rather than surrounding review context alone. The strongest domain-specific negative coefficient of the interpretable classifier is the token *cancel* — complementary, not independently confirmatory, evidence for the same billing signal the aspect pipeline surfaces. All findings are descriptive and associative, not causal.

Contents

1. Introduction

Every day, millions of customers leave reviews on platforms such as Amazon, Yelp and Google Maps. Each review couples a structured signal — an overall star rating — with a block of free text explaining the reasons behind it. For a company, this is a continuous stream of customer feedback that requires no surveys; the difficulty is reading it at scale. *Sentiment analysis*, the task of inferring opinion polarity from text (Pang and Lee, 2008), makes this tractable, and *aspect-based sentiment analysis* (ABSA) refines it by attributing sentiment to specific facets such as delivery, price or quality (Pontiki et al., 2014).

This paper deliberately restricts itself to *classical* natural language processing and machine learning — bag-of-words features and linear or tree-based classifiers — without any large language model. This is not only a constraint but a research lens: classical models are cheap, reproducible and, above all, *interpretable* — here meaning coefficient-based, global feature-level interpretability (which tokens drive a prediction overall), not instance-level explanation or formally verified faithfulness — and the central question of this work is how much predictive performance, if any, that interpretability costs. We study two Amazon review categories from the 2023 release (Hou et al., 2024) and pursue one overarching and two specific research questions.

Main RQ
: To what extent can classical NLP and machine-learning methods extract review-level sentiment and product-related aspects from Amazon reviews, and what trade-offs between predictive performance, interpretability, and business usefulness emerge?

RQ1
: How do Naive Bayes, Logistic Regression, and Gradient Boosting compare in document-level sentiment classification, and what do their systematic errors — particularly for 3-star reviews — reveal about the limitations of bag-of-words representations?

RQ2
: Which product aspects are most frequently mentioned, and how do their sentence-level sentiment profiles differ across the two selected categories?

Contributions. (i) A reproducible classical pipeline (cleaning, TF-IDF features, three classifier families, evaluation) with an honest treatment of class imbalance, showing that accuracy is the wrong headline metric and that a balanced Logistic Regression performs strongest in the observed TF-IDF setting while staying interpretable. (ii) An unsupervised, sentence-level ABSA pipeline using a domain-adapted aspect lexicon and VADER, with a robustness check against the document classifier. (iii) A cross-method finding: two complementary analysis stages converge on billing-related dissatisfaction as a recurring subscription-product signal.

2. Background and Related Work

2.1. Sentiment analysis

Opinion mining and sentiment analysis are surveyed comprehensively by Pang and Lee (2008), who frame polarity classification as a supervised learning problem over textual features and discuss the role of domain, negation and subjectivity. A long line of work treats document-level polarity as text classification with bag-of-words or n-gram features (Maas et al., 2011; Manning et al., 2008); we follow this classical recipe rather than dense neural representations. Large-scale modelling of Amazon-style review text and star ratings was studied early by McAuley and Leskovec (2013), who jointly model latent rating dimensions and review text; the Amazon Reviews 2023 corpus used here (Hou et al., 2024) is a much larger, more recent successor dataset in the same lineage, but the present paper deliberately returns to a classical, fully interpretable pipeline rather than the latent or neural representations of that later work.

2.2. Aspect-based sentiment analysis

Pontiki et al. (2014) formalise ABSA in SemEval-2014 Task 4, distinguishing aspect-term extraction, aspect-category detection and per-aspect polarity, and provide gold-annotated benchmarks. Their setting is supervised; in the absence of aspect-level annotations for the Amazon categories studied here, we approximate the same goals with an unsupervised lexicon-plus-VADER approach (Section 4.8), which trades annotation cost for transparency. Taboada et al. (2011) survey classical, lexicon-based sentiment methods in detail and document their main weakness directly relevant here: general-purpose lexicons are not tuned to a specific domain, so words can carry different valence in context than in isolation (Section 6.7 checks this concern directly for our billing lexicon).

2.3. Imbalance, boosting, deployment-realistic evaluation and uncertainty

Four further strands inform our methodology. He and Garcia (2009) survey class imbalance and the metrics that remain informative under skew, motivating macro-F1 and per-class recall over plain accuracy throughout. For handling imbalance, Chawla et al. (2002) introduce SMOTE (synthetic minority oversampling); we avoid it, since interpolating synthetic points in a sparse, high-dimensional TF-IDF space is methodologically debatable, and instead use class-weighted loss functions and random undersampling, reporting the trade-off transparently. Friedman (2001) introduce gradient boosting as iterative fitting to the functional gradient of a loss function; scikit-learn's gradient-boosting implementation (Pedregosa et al., 2011) is our tree-based comparison point. On evaluation realism, Sculley et al. (2015) document how evaluation can diverge from deployment conditions (e.g. training/serving skew); since review volume here is concentrated in recent years, we use a strict chronological split and add a further temporal validation slice (Section 4.3) so that feature and hyperparameter selection never touches the final evaluation data. Finally, a point estimate on one finite test set says nothing about its sampling variability; we follow Efron and Tibshirani (1993) and report non-parametric bootstrap confidence intervals throughout RQ1 and RQ2.

2.4. Features, models and the neural contrast

TF-IDF weighting over unigrams and bigrams is the standard sparse text representation (Manning et al., 2008). We compare three classifier families that span the bias–variance and interpretability spectrum: Multinomial Naive Bayes (a generative linear baseline), Logistic Regression (a discriminative linear model whose weights are directly readable), and gradient-boosted decision trees (scikit-learn's `GradientBoostingClassifier`). For sentence-level sentiment we use VADER (Hutto and Gilbert, 2014), a rule-based valence lexicon designed for short, informal text that needs no training data. All models are implemented with scikit-learn (Pedregosa et al., 2011), spaCy (Honnibal et al., 2020) and NLTK (Bird et al., 2009). Transformer sentence encoders such as Sentence-BERT (Reimers and

Gurevych, 2019) achieve strong results on semantic tasks but are comparatively opaque and resource-intensive; for an operational feedback dashboard, the value of a model that can answer "*which words drove this prediction?*" is high, so this paper quantifies what is given up by staying classical and finds (Section 7) that the interpretable linear model is not outperformed by the more complex tree ensemble, even in a matched-sample comparison that removes any training-data-volume confound (Section 5.3).

2.5. Positioning: careful empirical application and evaluation

Most work above either compares classical text-classification methods without a deployment-realistic, leakage-free temporal evaluation (Maas et al., 2011), or formalises ABSA under a supervised, gold-annotated setting unavailable for arbitrary Amazon categories (Pontiki et al., 2014). This paper combines a classical, interpretable model comparison with validation-only selection and bootstrap uncertainty; a chronological holdout that is leakage-free *within each category* (Section 4.3); an explicit bridge between document-level classification and sentence-level, lexicon-based aspect sentiment on the *same* corpus (including a clause-level sensitivity check); and two small, under-studied subscription categories rather than a large, well-trodden one.

3. Dataset and Exploratory Analysis

3.1. Dataset and category choice

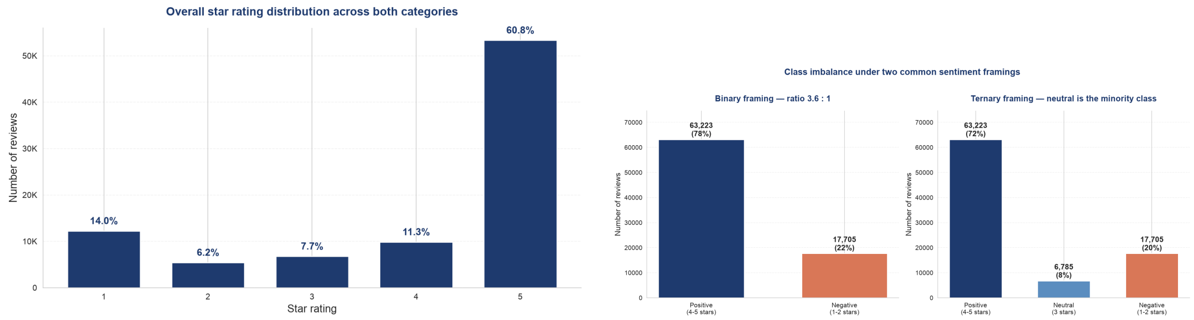
We use the Amazon Reviews 2023 dataset (Hou et al., 2024). The task asks for the smallest category; because RQ2 requires a *cross-category* comparison, we deliberately analyse *two* categories: **Subscription Boxes** (16,216 reviews, 641 distinct items — among the smallest categories) and **Magazine Subscriptions** (71,497 reviews), for a combined **87,713** reviews. Each review provides a 1–5 star `rating`, a `title`, free-text `body`, a `timestamp`, a `verified_purchase` flag and a `helpful_vote` count. Table 1 summarises the key statistics.

Table 1: Key dataset statistics from the exploratory analysis.

Statistic	Value
Total reviews	87,713
Subscription Boxes	16,216
Magazine Subscriptions	71,497
Time span	2001–2023
Verified-purchase share	82.7 %
Mean / median review length	40 / 22 words
5-star share	60.8 %
1-star share	14.0 %
Binary imbalance (pos:neg)	3.57 : 1
Neutral (3-star) share	7.7 %
Sample vocabulary with corpus frequency ≥ 5	6,308 terms

3.2. Star distribution and class imbalance

The rating distribution is strongly positive-skewed: 60.8% of reviews are 5-star and only 14.0% are 1-star (Figure 1). Mapping ratings to sentiment (Section 4.1) yields a 3.57 : 1 positive-to-negative imbalance in the binary framing and a small 7.7% neutral class in the ternary framing (Figure 2). This imbalance is the single most important property for evaluation: a classifier that always predicts "positive" already attains high accuracy while being useless, which is why we report macro-F1 and per-class recall throughout.



(a) Overall star distribution. **(b)** Sentiment-class imbalance.

Figure 1: Star ratings are heavily skewed toward 5 stars, producing a substantial positive/negative imbalance and a small neutral class.

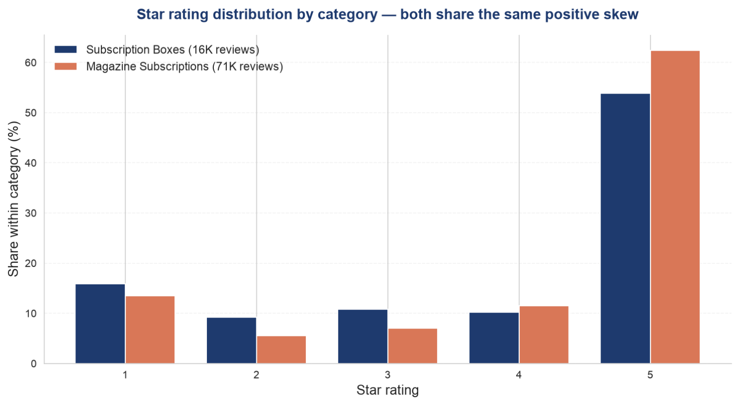
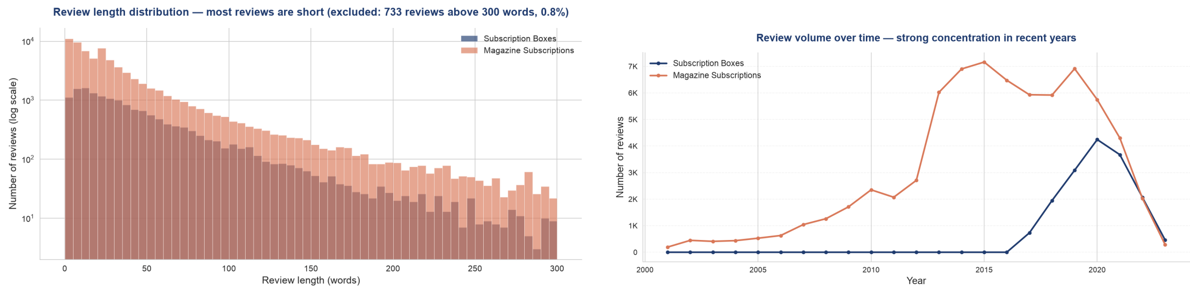


Figure 2: Per-category star distribution: both categories share the positive skew, confirming the imbalance is not an artefact of one category.

3.3. Review length and vocabulary

Reviews are short: a median of 22 words and a mean of 40 (Figure 3). Short texts carry few features, which both favours sparse linear models and motivates the sentence-level treatment in RQ2. A whitespace-tokenised vocabulary on a 20K-review sample contains 6,308 terms with sample corpus frequency at least five. This is a descriptive corpus statistic; the actual TF-IDF unigram feature count on the training split is reported separately in Section 5.



(a) Review-length distribution (words).

(b) Review volume over time.

Figure 3: Most reviews are short, and review volume is concentrated in recent years — which motivates a chronological rather than random train/test split.

3.4. Discriminative terms

A weighted log-odds analysis with an informative Dirichlet prior (Monroe et al., 2008) identifies the terms most associated with each sentiment class. Rendered as word clouds (Figure 4), the positive class is dominated by generic affect (*love, great, enjoy, wonderful*), while the negative class is strikingly concrete: *cancel, refund, subscription, ad, advertisement, charge, renewal*. Already at the descriptive stage, the data points to billing and advertising as the negative themes that the modelling later confirms.

Figure 4: Most discriminative terms per sentiment class (size ∝ weighted log-odds z-score). The negative cloud is a billing/advertising cluster.

4. Methodology

4.1. Sentiment labelling

Star ratings are the natural supervision signal but exist only per review, so RQ1 operates at the document level. We use two framings. The **binary** framing maps 1–2 stars to *negative*, 4–5 to *positive* and drops 3-star reviews; the **ternary** framing keeps 3 stars as *neutral*. Binary leads the analysis; ternary is reported to probe the hard, mixed-sentiment middle.

4.2. Text preprocessing

The body text is lowercased and stripped of HTML, URLs and non-alphabetic characters, then tokenised, stopword-filtered and lemmatised with spaCy (`en_core_web_sm`). A small number of reviews (238) reduce to an empty string and are dropped before modelling, as they carry no bag-of-words signal. The cleaned text feeds only the TF-IDF classifiers of RQ1; VADER in RQ2 deliberately consumes the *raw* text, because its valence rules exploit capitalisation, punctuation and intensifiers.

4.3. Chronological split without test-set leakage

Because review volume is concentrated in recent years, a random split would let a model train on future reviews (Sculley et al., 2015). We therefore sort by timestamp and split *within each category* so both categories appear on every slice, preserving the cross-category comparison RQ2 needs. Chronological order is therefore preserved *within each category*; because the per-category splits are then combined into one dev-train/validation/test set for the joint classifier, the resulting combined train and test periods partially overlap across categories (e.g. the combined training set can contain Magazine Subscriptions reviews newer than some Subscription Boxes reviews in the combined test set). The leakage-free, temporal guarantee this split provides is therefore *within-category*, not global: no review trains a model on data from its own category's future, but the combined classifier is not guaranteed to be evaluated on a period strictly later than all of its combined training data across categories.

A plain oldest-80%/newest-20% split is not enough on its own: if the held-out 20% is inspected repeatedly while comparing features, imbalance handling and hyperparameters, the final reported number is no longer an honest estimate of out-of-sample performance. We therefore split further into oldest **64% (dev-train)**, next **16% (validation)**, and newest **20% (final test)**. Feature representation (unigram vs. uni+bigram), imbalance handling, and a small deterministic hyperparameter grid — Naive Bayes $\alpha \in \{0.1, 0.5, 1.0, 2.0\}$ crossed with {no handling, reproducible undersampling}; Logistic Regression $C \in \{0.1, 0.5, 1.0, 2.0, 5.0\} \times \text{class\_weight} \in \{\text{none}, \text{balanced}\}$; Gradient Boosting $n_{\text{estimators}} \in \{40, 80\} \times \text{learning\_rate} \in \{0.05, 0.1\} \times \text{max\_depth} \in \{2, 3\} \times \text{class\_weight} \in \{\text{none}, \text{balanced}\}$ — are all selected using *only* the validation slice, by validation macro-F1 (primary) with minority-class recall as a transparent tie-breaker. The selected configuration is then retrained on the full oldest 80% (dev-train + validation) and evaluated on the final 20%. Final-test *labels, predictions and performance metrics* are never used for model, feature or hyperparameter selection; they are computed exactly once, after every modelling choice above has been fixed. This is distinct from the descriptive exploratory analysis (Section 3), which inspects the full dataset including the test period and so, unlike the modelling decisions above, was not shielded from the test period — acceptable for descriptive EDA, but worth keeping separate from the leakage-free claim about model selection.

4.4. Features

We build TF-IDF vectors with sublinear term-frequency scaling, min_df = 5 and a vocabulary cap of 20,000. Two configurations are compared: unigrams, and unigrams+bigrams; the vectoriser is fit on the training split only, and the choice between them is made on the validation slice (Section 4.3), not on the final test set.

4.5. Models and imbalance handling

We evaluate a **majority-class baseline** (always predicts positive), **Multinomial Naive Bayes**, **Logistic Regression** and `sklearn.ensemble.GradientBoostingClassifier` (Friedman, 2001; Pedregosa et al., 2011). Imbalance is handled two ways: Logistic Regression uses `class_weight=balanced`; Gradient Boosting uses balanced sample weights in the fitted sample; Naive Bayes, which has no class-weight parameter, is trained on a randomly undersampled balanced set rather than SMOTE (Chawla et al., 2002), which we consider methodologically debatable in a sparse, high-dimensional TF-IDF space (Section 2.3). Each model is also run without handling so the effect of imbalance is directly visible. The selected Gradient-Boosting configuration is trained on a deterministic stratified sample of 12,000 reviews regardless of how much training data is available, for tractable reproduction; this cap, and its consequence for the fairness of any full-data comparison, is reported transparently (Section 5.3).

4.6. Fair model-family comparison

Because Gradient Boosting trains on at most 12,000 rows while Naive Bayes and Logistic Regression can use the full training pool, a naive full-data comparison would conflate *model-family capability* with *training-data volume*. We therefore report two distinct blocks on the final test set: a **matched-sample comparison**, in which all three families train on an identical, deterministically sampled 12,000-row training set, and a **full-data comparison** (Naive Bayes and Logistic Regression only — Gradient Boosting is excluded because its 12,000-row cap means a "full-data" run would not actually reflect more training data than the matched comparison). The matched-sample block equalises the final training rows; hyperparameter selection, however, was conducted under each model's operational training regime (full dev-train for Naive Bayes and Logistic Regression, the internal 12,000-row cap for Gradient Boosting). We do not quantify the effect of this asymmetry, so it is not used to support any claim about general model-family superiority.

4.7. Evaluation and uncertainty

We report accuracy (for reference), macro-averaged F1, weighted F1 and per-class recall, plus confusion matrices. Macro-F1 and minority-class recall are the primary metrics under imbalance. All final-test metrics are additionally reported with non-parametric bootstrap 95% confidence intervals (2,000 replications, fixed seed, percentile method) following Efron and Tibshirani (1993), stratified by true class for per-class recall. The unigram-vs-bigram comparison uses a *paired* bootstrap (same resampled indices for both feature configurations in every replication) so the reported difference and its CI are directly comparable; if the CI excludes zero we describe a stable difference under this holdout, and if it includes zero we describe only a small observed gain, not a confirmed or significant improvement. Error analysis ranks the most confidently wrong predictions to characterise the hardest reviews, using a rule-based exploratory categorisation rather than manual coding.

4.8. Aspect-based pipeline (RQ2)

A review-level label cannot express that one review praises content but criticises price, so RQ2 works below the review. Each review is split into sentences; aspects are detected per sentence using a curated keyword lexicon of eight aspects — *delivery, packaging, quality, price, content, customer service, billing* and *advertising*. The last three are domain-specific: EDA surfaced content, billing/renewal and advertising vocabulary prominently. To avoid circularity, the detection lexicon contains aspect nouns/actions rather than sentiment words, and it excludes broad product identifiers such as *box*, *magazine* and *subscription*. A complementary, lexicon-free noun-phrase view (spaCy noun chunks) provides data-driven aspect candidates. Each aspect-mentioning sentence is scored with VADER (Hutto and Gilbert, 2014), and scores are aggregated per review, per aspect and per category.

Why VADER rather than the RQ1 classifier. The RQ1 classifier is trained on whole reviews; applying it to single sentences is an out-of-distribution use, since sentences are far shorter with a different feature distribution. VADER needs no training and is built for short text, so it is the *primary* sentence-level scorer. The document classifier is retained only as a *robustness check*: we measure how often the two methods agree rather than assuming either is ground truth (Section 6.4).

Defining "high coverage" / "substantial". Prior drafts called aspects "substantial" or "high coverage" without a fixed rule. We define an aspect as **high-coverage within a category iff at least 5% of that category's reviews mention it**, and apply this single threshold consistently in the methodology, results, discussion and conclusion.

Clause-level sensitivity analysis. Sentence-level attribution cannot separate two aspects that co-occur in one sentence with mixed sentiment ("great content . . . but overpriced"): both inherit the sentence's single VADER score. As a sensitivity check on this limitation — not a replacement for the sentence-level primary method — we additionally split sentences conservatively into clauses on a semicolon or one of *but, however, although, though, yet, while, whereas*, and attribute each aspect only to the clause(s) that contain its keyword (Section 6.6).

Billing-lexicon sensitivity check. The billing keyword list contains words such as *cancel*, *refund* and *charged* that often already occur in problem-oriented contexts. We check (a) each billing keyword's own VADER valence in isolation, and (b) billing sentiment recomputed with the matched keyword masked out before scoring, compared against the unmasked score (Section 6.7). This is a leakage/lexicon-sensitivity check, not an alternative primary method.

Review-level bootstrap uncertainty. The aspect-sentence table has multiple rows per review (one per mentioned aspect, occasionally more), so rows are not independent observations. We therefore resample whole *reviews*, not individual rows, for bootstrap 95% CIs of mean aspect sentiment per category and of the cross-category difference (2,000 replications, fixed seed; Section 6.8).

5. Results: Document-Level Sentiment (RQ1)

All model, feature and hyperparameter decisions in this section were made on a held-out **validation** slice (16% of the data) and never on the final test set; after those decisions were fixed, the selected configuration — Logistic Regression, `class_weight=balanced`, uni+bigram features — was retrained on the full oldest 80% and evaluated on the final 20% test set (Section 4.3). The validation-only grid search confirmed this same configuration as the best performer in both framings, ahead of every Naive Bayes and Gradient Boosting configuration tried, *before* the final test set was inspected.

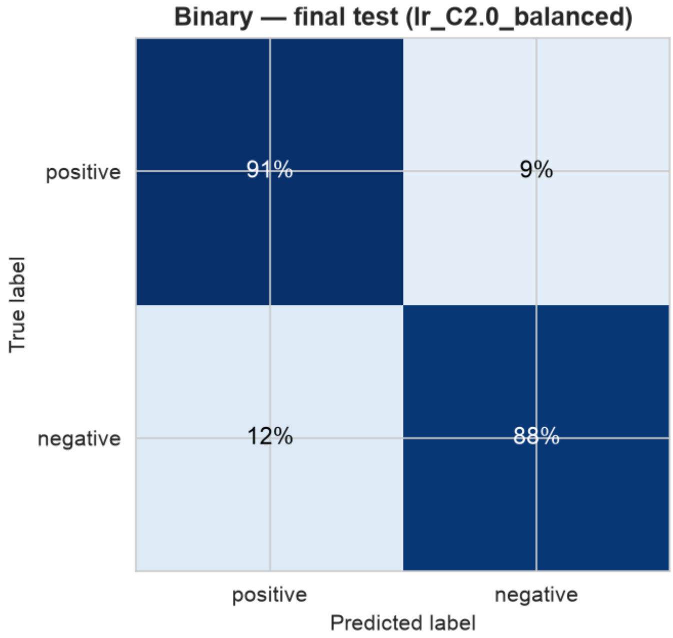
5.1. Binary and ternary framings

Table 2 (full 14-row grid in Appendix A) illustrates *why* imbalance handling matters, using the validation slice (not the final test set, and not used for any selection decision). In both framings the majority baseline has **zero** recall on the class(es) it never predicts; without handling, Naive Bayes' neutral recall collapses to 0.8% in the ternary framing, and Logistic Regression's negative/neutral recall stays low (0.69/0.12). Balancing or undersampling lifts minority recall substantially in both framings, at a real accuracy cost in the ternary case.

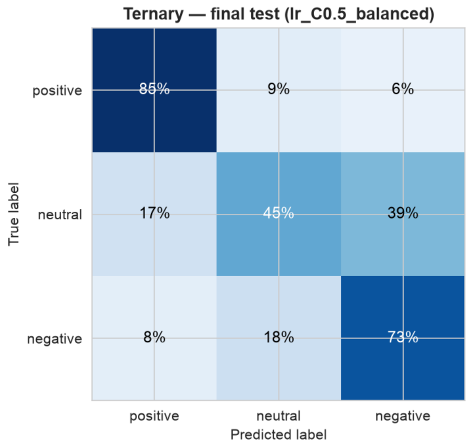
Table 2: RQ1 imbalance-handling demonstration, best and worst case per framing (validation slice, illustrative only; full 14-row grid in Appendix A).

Framing	Model	Acc.	Macro-F1	$\mathrm{Rec}_{\mathrm{pos}}$	$\mathrm{Rec}_{\mathrm{neu}}$	$\mathrm{Rec}_{\mathrm{neg}}$
Binary	Majority baseline	0.776	0.437	1.000	–	0.000
Binary	**Log. Reg. (balanced)**	**0.918**	**0.888**	0.926	–	0.889
Ternary	Majority baseline	0.719	0.279	1.000	0.000	0.000
Ternary	**Log. Reg. (balanced)**	**0.813**	**0.651**	0.880	0.449	0.712

On the final test set, accessed only after selection, the chosen binary configuration reaches accuracy 0.902, macro-F1 **0.888**, $\mathrm{recall}_{\mathrm{pos}} = 0.910$, $\mathrm{recall}_{\mathrm{neg}} = 0.884$; by category, macro-F1 is 0.894 for Magazine Subscriptions and 0.858 for Subscription Boxes (negative recall 0.881/0.903) — the combined classifier is not simply learning the larger category at the expense of the smaller one. The selected ternary configuration (C=0.5, balanced) reaches accuracy 0.785, macro-F1 **0.644**, $\mathrm{recall}_{\mathrm{pos}} = 0.848$, $\mathrm{recall}_{\mathrm{neu}} = 0.448$, $\mathrm{recall}_{\mathrm{neg}} = 0.731$ (95% bootstrap CIs for both framings in Table 8; confusion matrices in Figure 5). This is the central RQ1 trade-off: the 3-star class shows **persistent difficulty under the evaluated setup** on every held-out slice we tried (not a provable, universal ceiling) because such reviews genuinely mix positive and negative content — directly motivating both the dedicated 3-star analysis below (Section 5.7) and the sentence-level view of RQ2.



(a) Binary.



(b) Ternary.

Figure 5: Final test set confusion matrices, selected configuration. Neutral reviews (ternary) are most often confused with the adjacent positive class.

5.2. Statistical uncertainty

Table 8 reports 95% class-stratified bootstrap confidence intervals (2,000 replications, fixed seed) for every final-test metric above: each replication resamples within each true-class stratum at that stratum's original size (Section 4.3), so class proportions are fixed across replications rather than also varying by chance. For the unigram-vs-bigram comparison, both configurations were retrained on the full oldest 80% and evaluated on the final test set with a *paired*, equally stratified bootstrap (same resampled indices for both): macro-F1 is 0.875 (unigram) vs. 0.888 (uni+bigram), difference +0.013, 95% CI $[0.009, 0.016]$. Because this CI excludes zero, we describe this as a **stable, if modest, difference under this holdout** — not a "real" or independently significance-tested improvement, since no formal hypothesis test was run beyond the percentile bootstrap CI itself. The core CIs are: binary accuracy 0.902 $[0.897, 0.906]$, binary macro-F1 0.888 $[0.883, 0.893]$; ternary macro-F1 0.644 $[0.636, 0.652]$, ternary $\text{recall}_{\text{neu}}$ 0.448 $[0.421, 0.475]$. The full 9-row table (all metrics, both framings) is in Appendix A, Table 8.

5.3. Fair model-family comparison

Because the selected Gradient-Boosting configuration trains on at most 12,000 rows, a full-data comparison against Naive Bayes and Logistic Regression (trained on 64,456 rows) would conflate model-family capability with training-data volume. Table 3 reports two blocks on the final test set: a **matched-sample** comparison (all three families on an identical, deterministically sampled 12,000-row training set) and a **full-data** comparison (Naive Bayes and Logistic Regression only — Gradient Boosting is omitted because its internal 12,000-row cap means a "full-data" run would not reflect more training data than the matched comparison and would misrepresent the result as a scaling finding).

Table 3: Fair model-family comparison, binary framing, final test set. Each model uses its own validation-selected hyperparameters (Section 4.3); Naive Bayes' n_{train} in Block B is below 64,456 because its validation-selected configuration uses undersampling.

Block	Model	n_{train}	Accuracy	Macro-F1	$\text{Recall}_{\text{neg}}$
Matched (A)	Naive Bayes	12,000	0.855	0.814	0.616
Matched (A)	Logistic Regression	12,000	0.891	**0.876**	0.858
Matched (A)	`GradientBoostingClassifier`	12,000	0.841	0.815	0.750
Full data (B)	Naive Bayes	25,230	0.870	0.857	0.909
Full data (B)	Logistic Regression	64,456	0.902	**0.888**	0.884

Logistic Regression wins **even in the matched-sample block**, where every model family trains on identical data — so its advantage is not an artefact of Gradient Boosting receiving less training data, though the margin over Naive Bayes is narrower than a naive (non-validation-tuned) Naive Bayes baseline would suggest once Naive Bayes is also given its own validation-selected hyperparameters. Combined with Block B, this supports calling Logistic Regression the **strongest observed model under the evaluated setup**, but the unequal full-data training sizes between Logistic Regression and Gradient Boosting are never, by themselves, used as evidence that linear models are categorically superior to gradient boosting in general.

5.4. Feature representation

The choice between unigrams and uni+bigrams was made on the validation slice (Table 4): bigrams roughly double the vocabulary for a small observed macro-F1 gain in both framings. The final-test paired-bootstrap verdict for this gain is reported in Section 5.2 above. A vocabulary sanity check confirms that 1,338 negation-bearing unigram/bigram features (e.g. *not worth*, *absolutely not*) survive preprocessing and enter the final TF-IDF vocabulary.

Table 4: Feature comparison on the validation slice (Logistic Regression, balanced).

	Binary		Ternary	
Features	# Feat.	Macro-F1	# Feat.	Macro-F1
TF-IDF unigram	8,341	0.872	8,819	0.636
TF-IDF uni+bigram	20,000	0.888	20,000	0.651

5.5. Interpretability

A linear TF-IDF model exposes a signed weight per feature. Figure 6 visualises the strongest weights from the final selected binary model, with the exact coefficient table in Appendix Table 6. The positive weights are generic affect; the negative weights are dominated by negation cues and domain-specific terms such as *disappointed, cancel, disappointing, poor* and the bigram *cancel subscription*. The strongest overall negative coefficient is *not*, which confirms that preserving negation matters. Among the domain-specific terms, *cancel* is the strongest billing signal and the fifth-largest negative coefficient overall — a feature-level precursor to the RQ2 billing result, read as complementary rather than independently confirmatory evidence (Section 7).

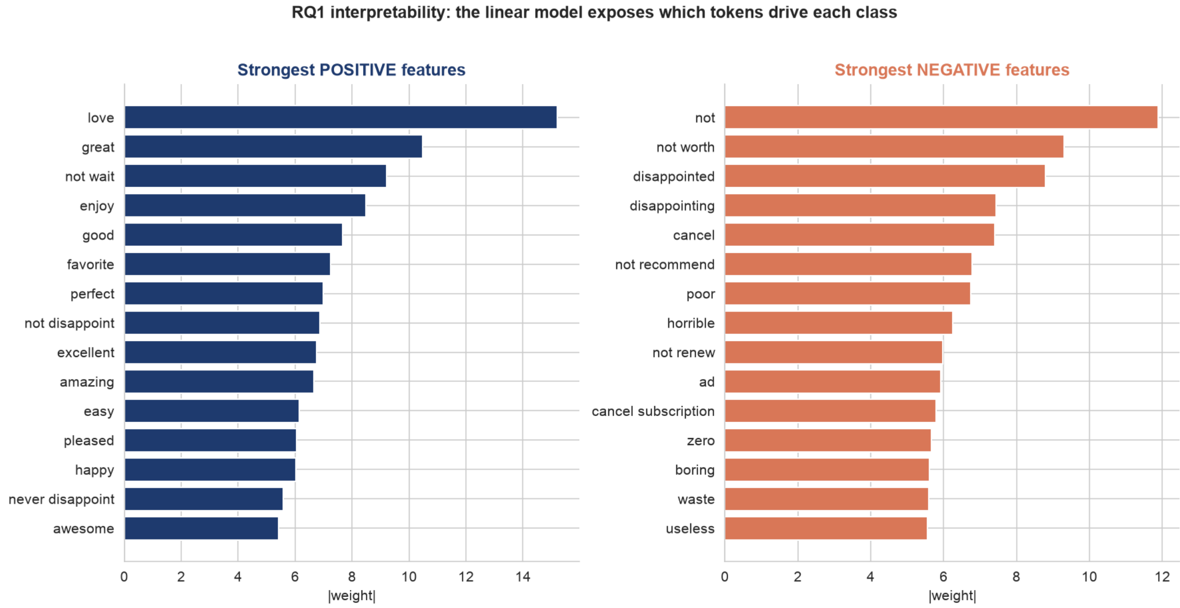


Figure 6: The linear model is fully auditable: each token carries an explicit, signed contribution to the positive/negative decision.

5.6. Error analysis: the hardest reviews

Ranking the most confidently wrong Logistic-Regression predictions (final test set) reveals that several errors appear to reflect disagreement between the rating and the review text rather than a simple preprocessing failure. To make this less anecdotal, the 100 highest-confidence binary errors were grouped with a **rule-based exploratory categorisation** (Appendix Table 9) — simple keyword/length heuristics written by the author, not a validated manual taxonomy and not human-coded data. The category named "potential rating–text disagreement / possible label noise candidate" deliberately avoids asserting that these cases are confirmed label noise. The heuristic suggests that such disagreement candidates form the largest group (47 of 100), followed by very short/context-free reviews (25) and mixed sentiment (14), but it is not a precise population estimate of how often label noise occurs in the dataset.

5.7. Direct analysis of 3-star reviews

The binary error analysis above never includes 3-star reviews (dropped under that framing). This section directly analyses all 1,248 reviews with true rating 3 in the final test set, using the selected ternary model. This is a **reproducible, automatic exploratory analysis** (contrastive-marker and lexical-cue word lists), not manual coding, and introduces no manual labels.

The model predicts neutral for 44.8% of true 3-star reviews (matching the ternary neutral recall above), negative for 38.5%, and positive for only 16.7% — the model's confusion skews toward negative, consistent with 3-star reviews often reading as mild complaints (Figure 7). Mean predicted-class confidence is similar across all three predicted classes (0.61–0.63) and barely differs between correct and incorrect predictions, so predicted probability alone is not a reliable signal for catching the model's mistakes here. Incorrect predictions are on average modestly longer than correct ones (43.2 vs. 40.6 words). A contrastive marker (*but/however/although/though/yet/while/whereas*) appears in 41.6% of 3-star reviews regardless of correctness, so contrastive markers alone do not strongly separate hard cases from easy ones in this automatic check; 18.4% of 3-star reviews contain both a positive and a negative lexical cue from fixed word lists, a direct automatic signature of mixed sentiment. Deterministically selected representative examples (the two highest-confidence 3-star reviews predicted into each class, not manually curated) make the pattern concrete: short enthusiastic 3-star reviews ("Love this!!!") get predicted positive; moderately worded genuine complaints ("The wording on this item was not clear at purchase... Chatted to cancel and surprisingly got a partial refund") get predicted negative; and explicitly lukewarm/mixed language ("Articles are somewhat interesting... Too many added advertisements") gets predicted neutral correctly. This directly answers the RQ1 sub-question on mixed-sentiment 3-star reviews: the limitation is not that the model is unconfident on these reviews, but that bag-of-words features cannot reliably separate mildly-positive from mildly-negative phrasing, and the star rating itself is sometimes only loosely tied to the text.

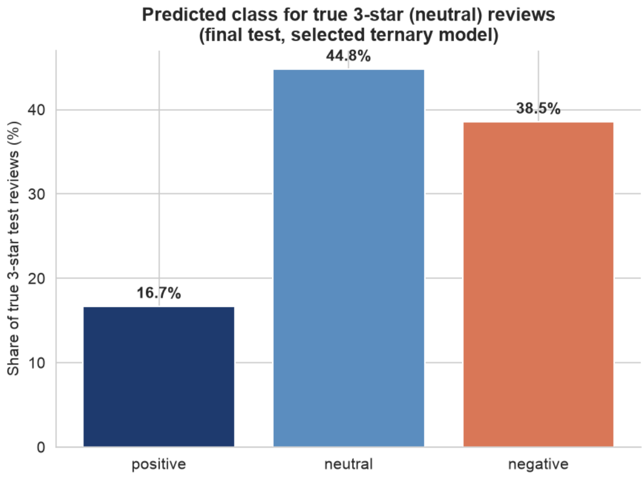


Figure 7: Predicted class for true 3-star (neutral) reviews, final test set, selected ternary model.

6. Results: Aspect-Based Sentiment (RQ2)

6.1. What customers discuss

Aspect mention rates differ sharply by category (Figure 8, Table 5). After cleaning the lexicon, *content* is the largest aspect in both categories: 43.0% of Subscription-Box reviews and 38.5% of Magazine reviews mention product or editorial substance. *Price* and *delivery* are secondary themes for boxes, while magazines show more *price* and *advertising* discussion. Crucially, *packaging* now appears in only 6.1% of box reviews; the earlier high value was a product-name artefact caused by matching *box/boxes*, not evidence that packaging dominates the category. Because Subscription-Box reviews are on average longer than Magazine reviews (Section 3), we also checked mentions per 100 words as a length-normalised sensitivity view; the ranking of aspects by discussion volume is materially unchanged (content remains the largest theme in both categories), so the per-review mention rate is retained as the headline statistic.

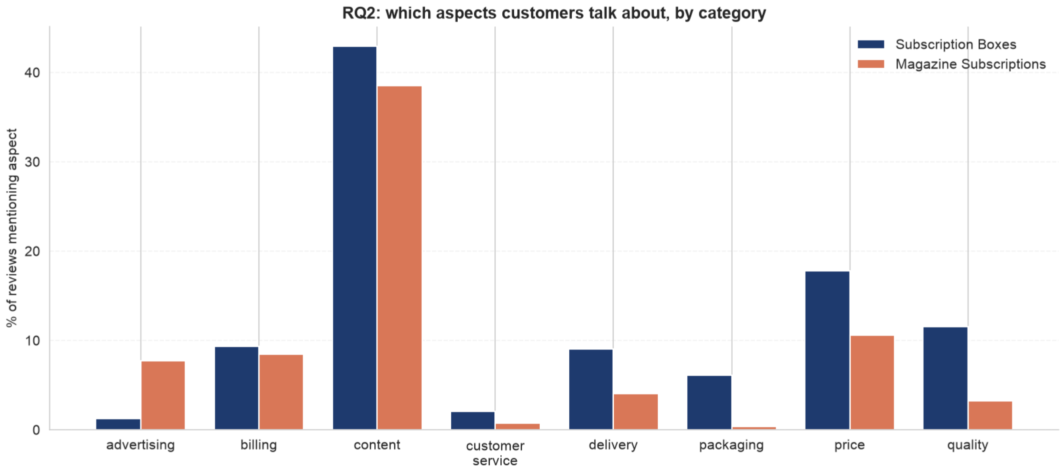


Figure 8: Share of reviews mentioning each aspect, by category. The cleaned lexicon shows content as the largest aspect and removes the earlier packaging artefact caused by product-name matches.

6.2. Aspect sentiment

The radar and bar views (Figures 9, 10) and Table 5 agree on one pattern: *billing* is high-coverage (mention rate $\geq 5\%$, Section 4.8) in *both* categories and is the **weakest high-coverage aspect in both** (compound -0.091 for boxes, 0.051 for magazines). For Subscription Boxes the raw lexicon score is the lowest (56.9% negative share), but the masking check below shows this polarity is not robust after removing trigger terms. For Magazine Subscriptions the mean is only slightly above neutral, so we do *not* describe billing there as an unconditionally negative pain point — only as the weakest aspect among the high-coverage ones. This is much sharper than the earlier broad-*subscription* version of the lexicon: once neutral product mentions are removed, billing's negative skew for boxes becomes visible rather than being diluted by neutral mentions. *Content* and *quality* are the most positive high-coverage aspects in boxes (compound ≈ 0.33–0.34), while magazines are most positive about *content* and *price* (compound ≈ 0.32–0.33). Advertising is high-coverage only in magazines (7.7% mention rate) and is not the dominant pain point there. The sentiment composition per aspect (Figure 10) shows the same pattern: billing carries the largest red (negative) band among high-coverage aspects. Customer service is numerically more negative in Magazine Subscriptions (-0.029) than billing there, but at a 0.8% mention rate (540 review-level detections, only 13 sentences in the separate VADER–classifier agreement subsample) it is below the high-coverage threshold and is treated as a low-coverage signal, not a primary finding.

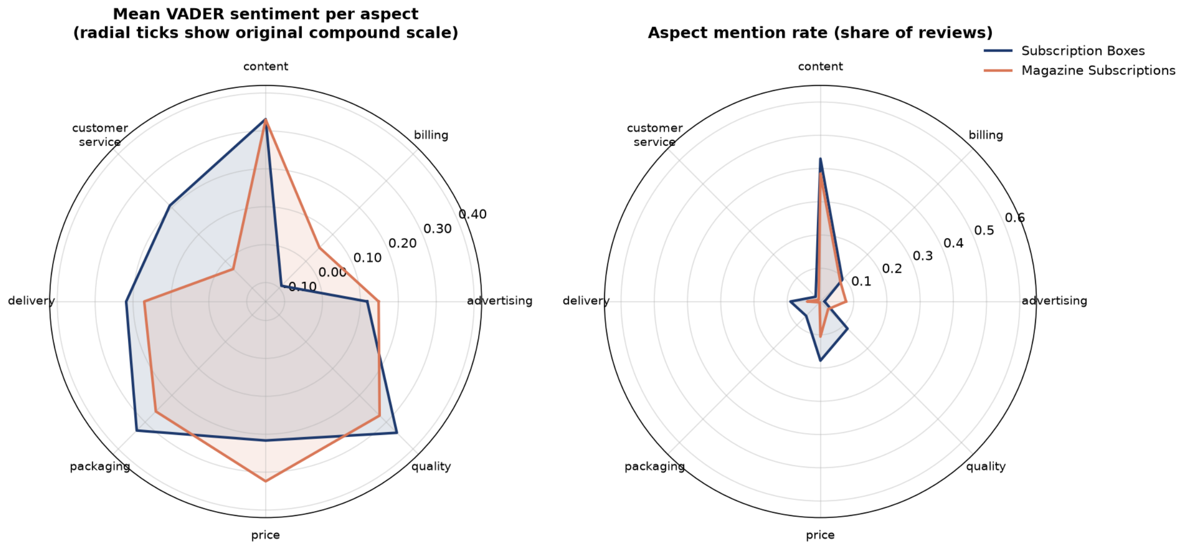
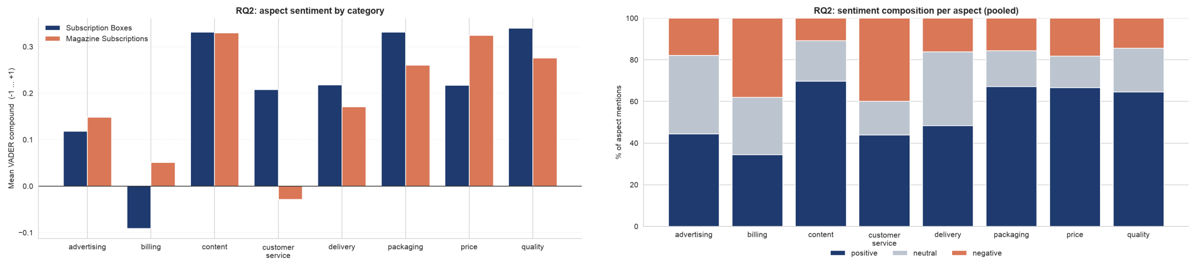


Figure 9: Aspect profiles as radar charts, with a dynamic, non-clipping radial range (see Section 4.8). Left: mean VADER sentiment, plotted via $(\text{compound} + 1)/2$ for polar geometry but labelled on the original $[-1, 1]$ scale — billing is the weakest high-coverage aspect in both categories, with low-coverage customer service a separate, lower-volume negative signal for magazines. Right: mention rate — content is common in both categories, while advertising is high-coverage only in magazines.

Table 5: Per-aspect mention rate, mean VADER compound and negative share, by category. ✓marks high-coverage aspects (mention rate $\geq 5\%$, Section 4.8); lowest sentiment among high-coverage aspects in **bold**.

	Subscription Boxes				Magazine Subscriptions			
Aspect	Mention %	Mean	Neg %	HC	Mention %	Mean	Neg %	HC
advertising	1.2	0.118	21.4		7.7	0.148	17.8	✓
billing	9.3	**-0.091**	56.9	✓	8.5	**0.051**	33.4	✓
content	43.0	0.331	12.9	✓	38.5	0.330	10.3	✓
customer service	2.0	0.208	27.7		0.8	-0.029	47.4	
delivery	9.0	0.218	20.7	✓	4.0	0.171	14.0	
packaging	6.1	0.331	15.8	✓	0.3	0.260	14.6	
price	17.8	0.217	25.0	✓	10.6	0.324	15.7	✓
quality	11.5	0.340	14.5	✓	3.3	0.275	14.4	



(a) Mean sentiment per aspect. **(b)** Sentiment composition (pooled).

Figure 10: Aspect sentiment as bars and as positive/neutral/negative composition. Billing is the weakest high-coverage aspect in both categories.

6.3. Noun-phrase view

The most frequent noun phrases (Figure 11) serve as a useful stress test rather than a blind validation of the lexicon. Many frequent phrases are product/category nouns — *the box*, *this magazine*, *the subscription* — and are intentionally *not* mapped to an aspect, because doing so would recreate the artefact the cleaned lexicon removes. As a simple keyword-vs.-noun-phrase check, 8 of the top 30 box phrases and 10 of the top 30 magazine phrases map back to at least one lexicon aspect. Matched phrases include *the toys*, *the items*, *the price*, *ads*, *articles* and *recipes*; the unmatched remainder is mostly generic language, category labels or product names. This makes noun phrases a discovery check, not a replacement for the curated lexicon.

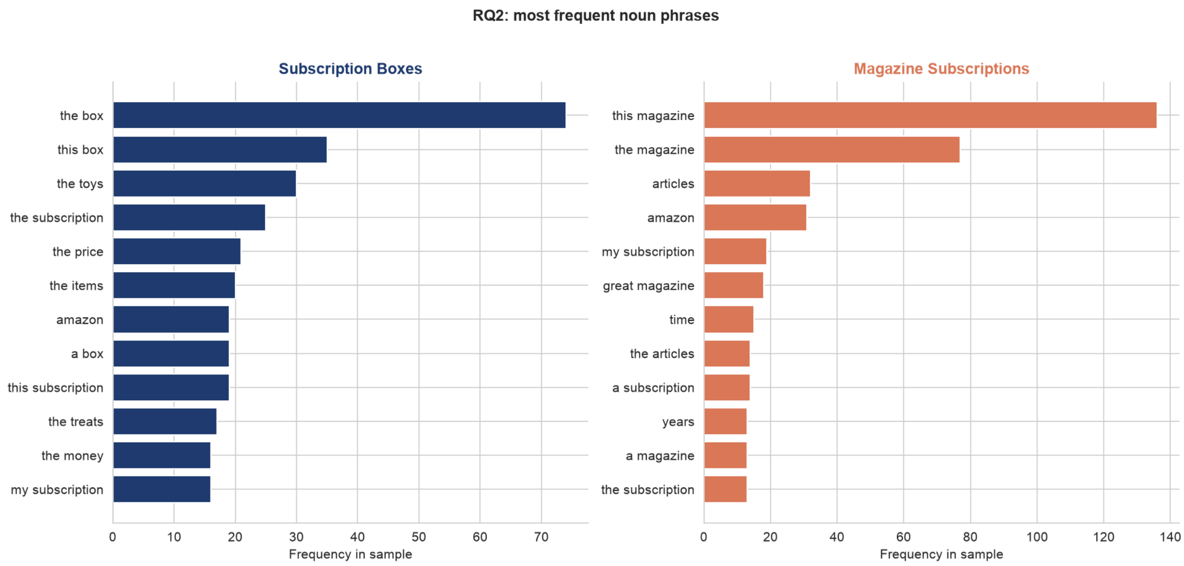


Figure 11: Most frequent noun phrases per category — a data-driven cross-check of the keyword aspect lexicon.

6.4. Robustness: VADER versus the classifier

On a test-set sample, VADER and the RQ1 classifier (final selected configuration, Section 4.3) agree on 77.1% of non-neutral aspect mentions ($n = 789$; Appendix Table 7). VADER labels 26.1% of sampled aspect mentions as *neutral*, a category the binary classifier structurally cannot express. The agreement provides a limited consistency check between two text-based methods, but cannot establish aspect-level validity in the absence of gold annotations. Combined with the out-of-distribution concern, this supports using VADER as the primary sentence-level scorer and the classifier as a sanity check only.

6.5. Illustrative qualitative spot-check of the lexicon

Because no gold aspect labels exist, we inspected a small, deterministic sample of 80 aspect-assigned sentences (ten per aspect; Appendix A). **This is not an independent or blinded manual annotation**: the inspection and the resulting judgement were carried out by the author and encoded directly in the notebook's source code, with no second annotator and no separately stored annotation file. Under a simple coding rule — a match is counted as plausible when the sentence explicitly discusses the detected product or service facet, not merely a category name or a sentiment word — 77 of 80 sampled assignments were judged plausible by the author. This is reported only as an **illustrative qualitative spot-check**, not as a validated precision estimate or a gold-standard evaluation; the sample is also too small to support a statistically reliable estimate even if it had been independently annotated. It does check the highest-risk failure modes directly: *box* no longer creates packaging hits, *subscription* no longer creates billing hits, and sentiment words no longer drive aspect detection. The remaining limitation is therefore mainly granularity — one sentence can still mention several aspects and receive one shared VADER score, which the clause-level sensitivity analysis below addresses as a robustness check.

6.6. Clause-level sensitivity analysis

Sentence-level attribution remains the primary method, but we directly check whether it drives the headline findings by re-scoring on a conservative clause split (a semicolon, or one of *but*, *however*, *although*, *though*, *yet*, *while*, *whereas*), attributing each aspect only to the clause containing its keyword, on a deterministic 8,000-review-per-category sample. The headline aspect *ranking* is essentially unchanged: the rank correlation between sentence-level and clause-level aspect-sentiment profiles is $\rho = 0.976$ (Subscription Boxes) and $\rho = 1.000$ (Magazine Subscriptions, $n = 8$ aspects each). Absolute scores shift slightly more negative at clause level for almost every aspect (Appendix Table 10); billing for Subscription Boxes moves from -0.080 (sentence) to -0.098 (clause) — *strengthening*, not weakening, the billing finding. Among reviews with at least one detected aspect, 17–22% have at least one aspect whose sentiment changes when scored from its own clause instead of the whole sentence; reviews without a detected aspect cannot be affected by this sensitivity check. A real example shows why this matters: "This was my first subscription box and I really enjoy it. I had a little problem with an item that was damaged during shipping, but the customer service was super nice about it and sent out a replacement right away." At sentence level, *content* (triggered by *item*) inherits the whole second sentence's positive compound (0.812, dominated by "super nice", "replacement right away"); at clause level, *content* is correctly scored from its own clause alone (-0.649), while *customer_service* keeps the positive clause. The main RQ2 findings are not revised by this analysis, since it does not reverse them — it mildly reinforces the billing result.

6.7. Billing-lexicon sensitivity check

Several billing keywords carry negative VADER valence in isolation: *cancel*/*cancelled* score -0.250, *charged* scores -0.202, and *charges* scores -0.273, while most others (*refund*, *billing*, *renewal*, ...)

score 0.0 alone. We therefore recompute billing sentiment with the matched keyword masked out before VADER scoring and compare to the unmasked score on the same sentences. Masking shifts mean sentiment notably toward positive in *both* categories: Magazine Subscriptions $0.036 \to 0.085$ ($+0.049$); Subscription Boxes $-0.080 \to +0.044$ ($+0.124$, **reversing the sign**). This is an important, honest caveat: a meaningful part of the unmasked billing-negative finding, especially for Subscription Boxes, reflects the lexicon valence of words like *cancel* and *charged* themselves, not solely the surrounding review context. We do *not* adopt the masked variant as a new primary method — masking also strips genuine sentiment-bearing context in many sentences (e.g. "terrible — they keep charging me after I cancelled", where *terrible* and the surrounding clause carry sentiment that survives masking) — but report it as a transparent limitation on how strongly the unmasked billing finding should be read as purely context-driven.

6.8. Uncertainty: review-level bootstrap

Because several aspect-sentence rows can come from the same review, we resample whole *reviews* (not rows) for 95% bootstrap CIs of mean aspect sentiment per category and of the cross-category difference (2,000 replications, fixed seed; Appendix Table 11). The underlying code and CSV name the difference column after the actual category names (Magazine Subscriptions minus Subscription Boxes), so the direction cannot be misread from an abstract "a"/"b" convention. The billing difference (Magazine − Boxes, compound) is $+0.142$, 95% CI $[0.122, 0.161]$ — excluding zero, so we describe a stable cross-category difference under this sample. Price is also higher for Magazine than Boxes ($+0.108$, CI $[0.091, 0.126]$, excludes zero), while quality is higher for Boxes than Magazine (-0.064 Magazine-minus-Boxes, CI $[-0.089, -0.039]$, excludes zero). Content's cross-category difference CI ($[-0.010, 0.008]$) includes zero, so we do *not* claim a stable difference there.

7. Discussion

Performance versus interpretability (Main RQ). Logistic Regression is the **strongest observed model under the evaluated setup** without sacrificing feature-level interpretability. The balanced binary model reaches macro-F1 0.888 on the final test set, ahead of Gradient Boosting in *both* the matched-sample comparison (0.876 vs. 0.815, identical 12,000-row training set, Section 5.3) and the full-data comparison against Naive Bayes (0.888 vs. 0.857, each with its own validation-selected hyperparameters). Because Logistic Regression wins even when every model family sees identical training data, this conclusion does not rest on Gradient Boosting having been given less data, and we do not extrapolate it into a general claim that linear models are categorically superior to gradient boosting outside this evaluated setup.

Complementary evidence for billing. The strongest negative coefficient of the supervised classifier is the general negation cue *not*; among domain-specific features, *cancel* is the strongest and the fifth-largest negative coefficient overall. The sentence-level VADER pipeline also identifies *billing* as the weakest high-coverage aspect in both categories. These two analyses provide **complementary, triangulating evidence** for billing-related dissatisfaction as a recurring subscription-product signal, rather than one independently confirming the other — both ultimately rely on overlapping billing-related vocabulary. The billing-lexicon masking check (Section 6.7) makes this connection explicit and shows its limit: masking the matched keyword shifts billing sentiment notably toward positive in both categories (reversing the sign for Subscription Boxes), so part of the billing-negative finding is the lexicon's own valence rather than purely the surrounding review context.

Label noise candidates and the neutral class. Several of the highest-confidence errors appear to reflect disagreement between the rating and the review text, and the ternary results showed that the 3-star class cannot be reliably separated under this bag-of-words setup. Both point to the same limitation: the star rating is a noisy, coarse label, and many of the hardest reviews are mixed or context-poor — exactly where aspect-level analysis can add value over a single document label. The dedicated 3-star analysis (Section 5.7) shows this is consistent with a representational limitation rather than a confidence problem: predicted-class confidence alone does not reliably separate correct from incorrect predictions here, and bag-of-words features cannot reliably tell mildly-positive from mildly-negative phrasing apart.

Business implications. A practitioner can deploy the balanced Logistic Regression as a transparent polarity monitor, and the lexicon+VADER layer as an aspect dashboard that flags *which* facet drives sentiment. Billing is the **weakest high-coverage aspect in both categories**; complaint language around it is frequent and scores negative under the lexicon, but the billing-lexicon masking check (Section 6.7) shows this negative polarity is **not robust for Subscription Boxes** once the triggering terms are masked out, so for Boxes the aspect's value lies primarily in its high mention rate and refund/cancel volume rather than in a confidently negative sentiment score (Magazine Subscriptions is only slightly positive on billing, so not an unconditional issue there either). For Boxes specifically, the operational response is still to review renewal, cancellation and refund flows, monitoring billing negative share and refund/cancel mention rate. Advertising is a magazine-specific signal worth tracking alongside churn; boxes otherwise need closer monitoring of content, price and delivery. Difficult 3-star reviews are better treated as candidates for aspect-level review than forced into a single polarity label. Because both categories are subscription products (Section 8), these recommendations need re-validation before extending to non-subscription Amazon categories.

8. Limitations and Future Work

- **Sentence-level attribution** is the primary method and cannot, by itself, separate two aspects that co-occur in one sentence with mixed sentiment ("great content … but overpriced"). The clause-level sensitivity analysis (Section 6.6) probes this directly, but remains a sensitivity check, not a redesigned primary method.
- **Lexicon and scorer** are hand-built and English-only, and VADER is general-purpose rather than domain-tuned (Taboada et al., 2011); the billing-lexicon masking check (Section 6.7) is a partial, not exhaustive, probe of this concern. The *content* lexicon in particular includes broad terms (*item, product, issue*) so that it can match generic mentions of what is being reviewed; for Magazine Subscriptions *issue* is ambiguous between "magazine issue" and "problem", so part of content's high coverage there reflects lexicon breadth rather than substantive discussion alone.
- **No gold aspect labels** exist, so RQ2 is validated only against a second method and an *illustrative*, self-conducted qualitative spot-check (not an independent or blinded annotation), not ground truth. A small annotated set (SemEval-style, Pontiki et al., 2014) with a second, independent annotator would enable proper scoring.

- **Star ratings as labels** are noisy, as the exploratory error categorisation suggests (these are possible label-noise candidates, not confirmed label noise), placing a ceiling on supervised accuracy.
- **Hyperparameter and feature search** used a small, deterministic grid (Section 4.3), not an exhaustive search, to keep notebook runtime realistic for a seminar project; a larger search could shift the selected configuration.
- **Cross-category scope:** both categories studied are subscription products, so RQ2's cross-category comparison is within a fairly narrow product domain and should not be generalised to arbitrary Amazon categories without further checks. Subscription Boxes has only 641 distinct items, so its aspect frequencies can be influenced by a few popular boxes; the two categories also differ in average review length, which is why mention rates are also reported per 100 words as a sensitivity check.
- **Associative, not causal:** all RQ1 and RQ2 findings, including the billing connection across the two analyses, are descriptive and associative.
- **Neural contrast:** a transparent linear model was the goal here; a controlled comparison against a Sentence-BERT classifier (Reimers and Gurevych, 2019) would quantify the interpretability–accuracy frontier more precisely.

9. Conclusion

We built a classical NLP pipeline, reproducible given the specified dataset and environment, that turns 87,713 Amazon subscription reviews into structured opinion data. For document-level polarity, across the evaluated models, imbalance handling had a larger observed effect on minority-class performance than classifier choice: a 69%-accurate baseline identifies no negatives, and only after explicit rebalancing do the models recall the minority class well, with an interpretable balanced Logistic Regression the **strongest observed model** (0.888 macro-F1, final test) under the evaluated setup — supported by a matched-sample comparison that reduces the concern that the result is only a training-data-volume artefact, and reported with bootstrap confidence intervals rather than as bare point estimates. A direct analysis of true 3-star test reviews shows the ternary model's persistent difficulty under this setup is consistent with a representational limitation rather than a confidence problem, since confidence alone does not reliably separate correct from incorrect predictions. For aspect-based sentiment, a sentence-level lexicon+VADER pipeline (cross-checked with a clause-level sensitivity analysis, a billing-lexicon masking check, and review-level bootstrap CIs) shows that *what* customers discuss is mostly content, while *what frustrates* them is shared and concentrated in *billing* — the weakest high-coverage aspect in both categories. Billing complaint language is frequent and scores negative under the lexicon, but its polarity is not robust for Subscription Boxes specifically: the masking check shows the negative skew there largely depends on the triggering terms (*cancel*, *refund*, *charged*) themselves rather than on surrounding review context alone. The fact that the interpretable classifier's strongest domain-specific negative word is *cancel* — the same billing signal — means the two analyses provide complementary, triangulating evidence rather than independent confirmation, and all results here remain associative and descriptive, not causal, restricted to two subscription-product categories rather than Amazon categories in general.

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A. Additional material

The repository (`https://github.com/ivelinayna/seminar`) contains, under `results/tables/`: the full validation-slice imbalance-handling grid behind Table 2; per-class precision/recall/F1; the hyperparameter search, fair model-family, bootstrap CI, 3-star, clause-level/billing-lexicon sensitivity, and top-noun-phrase tables; and the sentences used for the lexicon spot-check (self-conducted, not independent or blinded). It also includes a `pytest` suite (`tests/`); all notebooks reproduce every figure and table from raw data — see `README.md` for commands, environment, and seeds.

Table 6: Strongest Logistic-Regression coefficients (binary, balanced, final selected model).

Positive term	Weight	Negative term	Weight
love	+15.2	not	−11.9
great	+10.5	not worth	−9.3
not wait	+9.2	disappointed	−8.8
enjoy	+8.5	disappointing	−7.4
good	+7.7	cancel	−7.4
favorite	+7.2	not recommend	−6.8
perfect	+7.0	poor	−6.7
not disappoint	+6.9	horrible	−6.2

Table 7: Agreement between VADER and the document classifier on non-neutral aspect mentions (test sample).

Aspect	Agreement	*n*
advertising	0.512	43
billing	0.758	95
content	0.796	392
customer service	0.769	13
delivery	0.844	45
packaging	0.933	15
price	0.735	132
quality	0.796	54
Overall	**0.771**	789

Table 8: Class-stratified bootstrap 95% CIs (2,000 replications), final test set.

Framing	Metric	Point	CI low	CI high
Binary	Accuracy	0.902	0.897	0.906
Binary	Macro-F1	0.888	0.883	0.893
Binary	Recall positive	0.910	0.904	0.915
Binary	Recall negative	0.884	0.875	0.893
Ternary	Accuracy	0.785	0.779	0.791
Ternary	Macro-F1	0.644	0.636	0.652
Ternary	Recall positive	0.848	0.841	0.854
Ternary	Recall neutral	0.448	0.421	0.475
Ternary	Recall negative	0.731	0.719	0.743

Table 9: Rule-based exploratory categorisation of the 100 highest-confidence binary Logistic-Regression errors (final test set).

Error category	Count	Share
Potential rating–text disagreement / possible label noise candidate	47	47.0 %
Very short or context-free review	25	25.0 %
Mixed sentiment	14	14.0 %
Domain ambiguity	8	8.0 %
Probable genuine model error	4	4.0 %
Aspect conflict	1	1.0 %
Negation	1	1.0 %

Table 10: Sentence-level vs. clause-level mean VADER compound, by aspect (8,000-review-per-category sample).

	Subscription Boxes			Magazine Subscriptions		
Aspect	Sent.	Clause	Diff	Sent.	Clause	Diff
advertising	0.076	0.063	-0.013	0.171	0.139	-0.032
billing	-0.080	-0.098	-0.018	0.036	0.020	-0.016
content	0.332	0.321	-0.011	0.322	0.307	-0.015
customer service	0.176	0.181	0.005	0.013	-0.001	-0.014
delivery	0.233	0.213	-0.021	0.179	0.146	-0.033
packaging	0.330	0.324	-0.006	0.252	0.241	-0.010
price	0.221	0.200	-0.021	0.316	0.303	-0.013
quality	0.348	0.341	-0.007	0.307	0.288	-0.019

Table 11: Review-level bootstrap 95% CIs for mean aspect sentiment (2,000 replications). Diff column is Magazine Subscriptions minus Subscription Boxes.

Aspect	Mean (Boxes) [CI]	Mean (Magazine) [CI]	Diff (Mag.−Boxes) [CI]
advertising	0.118 [0.059, 0.173]	0.148 [0.139, 0.157]	0.030 [-0.026, 0.088]
billing	-0.091 [-0.109, -0.074]	0.051 [0.042, 0.060]	0.142 [0.122, 0.161]
content	0.331 [0.322, 0.340]	0.330 [0.326, 0.334]	-0.001 [-0.010, 0.008]
customer service	0.208 [0.152, 0.262]	-0.029 [-0.065, 0.008]	-0.236 [-0.302, -0.172]
delivery	0.218 [0.196, 0.239]	0.171 [0.158, 0.184]	-0.048 [-0.073, -0.021]
packaging	0.331 [0.306, 0.357]	0.260 [0.213, 0.309]	-0.071 [-0.128, -0.017]
price	0.217 [0.201, 0.231]	0.324 [0.315, 0.334]	0.108 [0.091, 0.126]
quality	0.340 [0.321, 0.359]	0.275 [0.258, 0.292]	-0.064 [-0.089, -0.039]

Declaration of Originality

I declare that I have authored this paper independently, that I have not used other than the declared sources / resources, and that I have explicitly marked all material which has been quoted either literally or by content from the used sources.

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